# A278299: Smallest Complete Polyomino Coloring

Minimum cells in a polyomino with $n$-coloring where every color pair shares an edge

Proved: $a(12) = 40$, $a(13) = 46$, $a(14) = 56$, $a(15) = 61$, $a(16) = 69$, $a(17) = 77$, $a(18) = 90$. Upper bounds: $a(19)l = 97$, $a(20)l = 108$, $a(21)l = 119$

*Computed by Peter Exley, March 2026*

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$a(12) = 40$ **[PROVED]** $5 \times 8$ bounding box, 66 color pairs. Edge bound: $65 < 66$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| F | D | C | K | D | J | H | B |
| L | A | E | F | H | I | G | D |
| D | I | L | J | A | B | L | C |
| E | K | H | E | G | J | K | B |
| B | G | C | I | F | C | A | F |

$a(13) = 46$ **[PROVED]** $7 \times 7$ bounding box, 78 color pairs. Edge bound: $76 < 78$

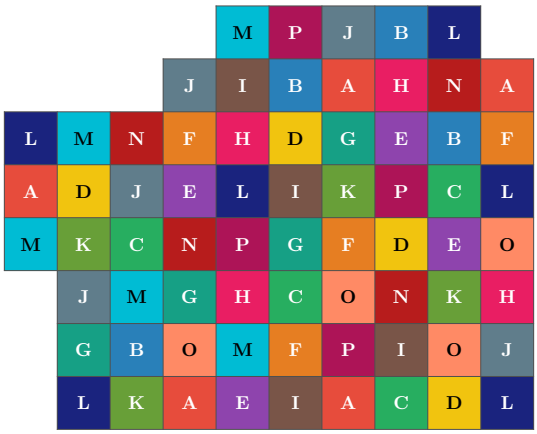
| | | | | | | |
|---|---|---|---|---|---|---|
| | A | J | L | M | J | |
| I | F | B | A | D | C | L |
| A | C | M | G | L | B | H |
| M | K | F | J | E | D | K |
| I | B | G | D | H | F | L |
| K | E | C | I | G | E | I |
| | M | H | J | K | A | H |

$a(14) = 56$ **[PROVED]** $7 \times 8$ bounding box, 91 color pairs. Contact bound: $12 < 13$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| G | F | L | B | A | D | F | B |
| M | E | G | N | M | L | C | M |
| J | B | I | J | K | A | H | F |
| E | K | D | M | I | F | K | N |
| I | C | B | H | L | J | G | C |
| K | A | G | D | E | A | H | J |
| L | N | E | C | H | I | N | D |

$a(15) = 61$ **[PROVED]** $7 \times 9$ bounding box, 105 color pairs. Edge bound: $104 < 105$

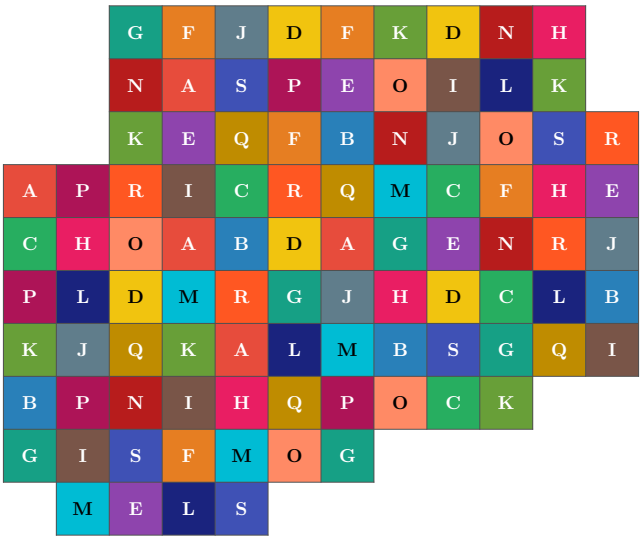
$a(16) = 69$ **[PROVED]** $8 \times 10$ bounding box, 120 color pairs. Edge bound: $119 < 120$



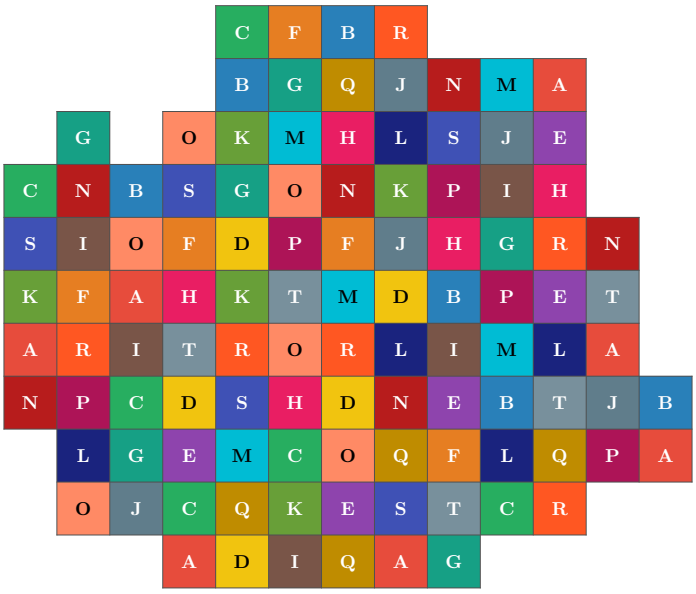

$a(17) = 77$ **[PROVED]** $8 \times 10$ bounding box, 136 color pairs. Edge bound: $134 < 136$

$a(18) = 90$ **[PROVED]** $9 \times 10$ bounding box, 153 color pairs. Contact bound: $16 < 17$

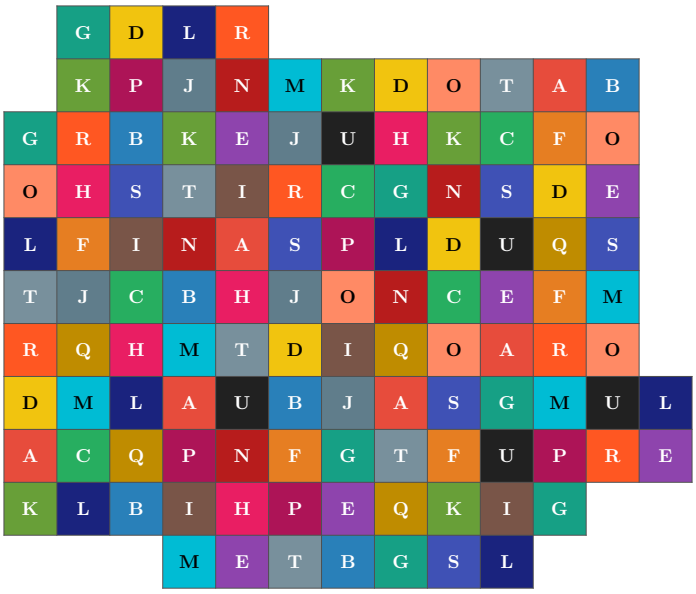
$a(19) = 97$ **[UPPER BOUND]** $10 \times 12$ bounding box, 171 color pairs. Computational upper bound




$a(20) = 108$ **[UPPER BOUND]** $11 \times 13$ bounding box, 190 color pairs. Computational upper bound



$a(21) = 119$ **[UPPER BOUND]** 11 × 13 bounding box, 210 color pairs. Computational upper bound

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**Color Legend**

A=0 B=1 C=2 D=3 E=4 F=5 G=6
H=7 I=8 J=9 K=10 L=11 M=12 N=13
O=14 P=15 Q=16 R=17 S=18 T=19 U=20